



UGANDA INDUSTRIAL RESEARCH INSTITUTE

Namanve Industrial and Business Park, Mukono District, Uganda

SYSTEM PROPOSAL

DESIGN AND DEVELOPMENT OF A WEB BASED

INVENTORY MANAGEMENT SYSTEM

FOR UGANDA INDUSTRIAL RESEARCH INSTITUTE (UIRI)

NAKAWA AND NAMANVE CAMPUSES

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Submitted to	ICT Team, UIRI Namanve
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Project Team and Submission Details

This proposal has been prepared as part of the ICT internship project at the Uganda Industrial Research Institute. It presents the planned database and web application approach for improving the tracking of inventory records across the UIRI working environment.

Student Name	Registration Number	Assigned Contribution
Keith Paul Kato	24/U/26593/EVE	Lead reporting, proposal preparation, database design and documentation coordination
Komukama Tracy	24/U/06151/EXT	Requirements review, user needs analysis and supporting documentation
Atuhiire Deo Kamate	24/U/14365/PS	System analysis, testing support and validation of database design decisions

Project Title	Design and Development of a Web Based Inventory Management System for UIRI
Project Type	Database and Web Application Development
Primary User Department	Information and Communications Technology (ICT)
Initial Pilot Site	UIRI Namanve
Planned Institutional Coverage	Nakawa and Namanve Campuses

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Abbreviations and Acronyms

Acronym	Full Form
DBMS	Database Management System
ERD	Entity Relationship Diagram
ICT	Information and Communications Technology
IMS	Inventory Management System
MMISDC	Machining, Manufacturing and Industrial Skills Development Centre
PDU	Procurement and Disposal Unit
RBAC	Role Based Access Control
SRS	Software Requirements Specification
UIRI	Uganda Industrial Research Institute

1.0 Introduction

This document presents a system proposal for the design and development of a web based Inventory Management System for the Uganda Industrial Research Institute. The proposed system is intended to provide a practical database driven platform for recording, tracking, monitoring and reporting inventory resources used within UIRI operations. The core system requirements are drawn from the UIRI ICT Team project brief [1].

The project was initiated after guidance from the ICT supervisor, Mr. William Kakooza, who identified the need for improved inventory tracking across UIRI. The system is expected to help the institute answer day to day inventory questions such as the number of computers in a department, the specification of a particular brand, the year of purchase, the acquisition batch, operational status, faulty items, serial numbers and current location [1].

The proposal therefore focuses on a system that can support both quantity based stock control and individual asset tracking. This is important because UIRI handles ordinary consumables as well as durable assets such as ICT equipment, laboratory instruments, engineering tools, machines and surveillance devices.

1.1 Purpose of the Proposal

The purpose of this proposal is to define the problem, justify the need for the system, identify the intended scope, describe the proposed approach and provide a foundation for the next deliverables. These next deliverables include the Software Requirements Specification, data dictionary, ER diagrams, UML diagrams, source code, user manual, final report and demonstration.

2.0 Background of the Organization

Uganda Industrial Research Institute is a government parastatal institution that supports industrial research, technology development, product development, machinery fabrication, value addition, business incubation and practical support to enterprises [2], [3]. The institution operates in a technical environment where accurate management of resources is central to smooth service delivery and institutional accountability.

UIRI has operational activities at Nakawa and Namanve. Nakawa hosts administrative, laboratory, product development, ICT and agro processing related functions. Namanve hosts the Machining, Manufacturing and Industrial Skills Development Centre, which is associated with engineering, fabrication, mechatronics, welding, CNC operations, wood technology, materials testing and industrial skills development [2], [4].

2.1 Inventory Context at UIRI

The inventory environment at UIRI is broad. It includes office assets, ICT devices, engineering tools, laboratory equipment, consumables, safety items and surveillance equipment. Such a setting requires a database design that separates fixed assets from consumable stock. Computers, machines and laboratory instruments must be traceable individually, while stationery, reagents, welding rods and other supplies must be monitored by quantity and reorder level.

Stakeholder Area	Representative Inventory Records Required
ICT Department	Computers, laptops, servers, routers, switches, projectors, printers, serial numbers and maintenance status
Stores and Administration	Office supplies, furniture, consumables, stock receipts, stock issues and general store balances
Research and Laboratories	Scientific equipment, testing materials, reagents, glassware and laboratory consumables
MMISDC Namanve	Machining tools, fabrication equipment, welding materials, CNC related items and safety equipment
Finance and Procurement	Supplier records, purchase references, asset values, acquisition batches and procurement related reports
CCTV and Surveillance	Cameras, network video recorders, monitors, security devices and their operational condition

3.0 Statement of the Problem

UIRI requires a more reliable way of tracking institutional inventory across its working departments and campuses. Where inventory information is kept in manual files, paper vouchers or separate spreadsheets, it becomes difficult to obtain a quick and dependable picture of what the institute owns, where each item is located and what condition it is in.

This creates several operational weaknesses. First, management may fail to know the real quantity of available items at a given time. Second, departments may request new items even when similar items already exist elsewhere. Third, faulty or retired equipment may remain recorded as available, creating ghost assets. Fourth, high value assets such as computers, machines and laboratory instruments may not be properly traced by serial number, brand, purchase year, batch and custodian.

The absence of a central database also affects reporting. Producing a report on all computers in Mechatronics, all faulty ICT equipment, all items supplied by a particular supplier or all stock issued in a given month becomes slow and prone to errors. There is therefore a justified need for a secure web based Inventory Management System that supports accurate recording, stock movement history, reports, user accountability and management decision making.

4.0 Literature Review

4.1 Inventory Management in Institutional Settings

Inventory management involves the control of items received, stored, issued, maintained, transferred or disposed of by an organization. In an industrial research institution, inventory management is broader than ordinary retail stock control because the same system must support consumables and durable assets. Consumables are reduced through use, while durable assets continue to exist and must be tracked by identity, condition, location and responsibility.

A proper database therefore needs a catalogue of item types, records of physical asset instances, stock balances and stock transactions. Treating everything as one simple quantity table would not be enough for UIRI because a computer with a serial number is not managed in the same way as a box of printer paper.

4.2 Review of Existing Inventory Control Approaches

Approach	Strength	Weakness
Manual ledgers	Simple to start and easy for staff already used to paper records	Slow reporting, weak search ability, limited audit trail and high risk of misplaced records
Spreadsheets	Cheap, familiar and useful for small lists	Files can diverge between departments and changes may not be properly controlled
Large ERP inventory modules	Strong functionality and broad enterprise coverage	May be costly, complex and too large for the immediate internship prototype
Custom web-based database system	Can be designed around UIRI workflows and hosted on the local intranet	Requires careful design, testing and internal maintenance after deployment

4.3 Identified Gap

The identified gap is the need for a localized inventory system that supports both asset tracking and stock control in one database. The system must be able to report on departments, campuses, suppliers, stock movement, low stock levels, asset condition, maintenance records and user activities. This gap is the basis for the proposed UIRI Inventory Management System.

5.0 Project Objectives

5.1 General Objective

The general objective of the project is to design and develop a secure web based Inventory Management System for Uganda Industrial Research Institute to support centralized tracking, control and reporting of inventory records across the Nakawa and Namanve campuses.

5.2 Specific Objectives

1. To study the current inventory tracking problem and identify the users, records, reports and business rules required by UIRI.
2. To design a normalized relational database that can store users, roles, campuses, departments, suppliers, item categories, inventory items, assets, stock balances, transactions and audit logs.
3. To implement secure user authentication and role based access control for administrators, store managers and staff users.
4. To enable registration, editing, classification and tracking of inventory items, including asset details such as brand, serial number, purchase year, batch and condition where applicable.
5. To support stock received, stock issued, stock transferred, stock returned and low stock alerts.
6. To provide supplier management and supplier transaction history.
7. To generate dashboard summaries and printable reports that support stakeholder decision making.
8. To enforce input validation, password protection, session control and audit logging of important user actions.

6.0 Scope of the Proposed System

6.1 Functional Scope

The system will support user registration, login, role based access, inventory item management, category management, supplier management, stock receipt, stock issue, stock transfer, stock return, low stock alerts, search, filtering, reporting and audit logging.

6.2 Organizational and Geographical Scope

The database structure is designed to support both Nakawa and Namanve campuses. The initial implementation may be piloted at UIRI Namanve because that is where the internship supervision and immediate project work are based. However, the campus and department tables will allow extension to Nakawa without redesigning the database.

6.3 Inventory Categories Covered

Category	Examples
ICT Equipment	Computers, laptops, printers, servers, routers, switches, projectors and monitors
Laboratory Equipment	Scientific instruments, testing equipment, reagents, glassware and laboratory supplies
Engineering and Industrial Assets	Welding machines, machining tools, CNC related tools, fabrication equipment and maintenance tools
Office Assets	Desks, chairs, cabinets, filing equipment, communication devices and general office equipment
Consumables	Stationery, printer cartridges, packaging materials, workshop materials and safety supplies
Safety and Surveillance Equipment	CCTV cameras, network video recorders, monitors and protective gear

6.4 Scope Exclusions

The proposed system will not implement payroll, full accounting, complete procurement approval workflows, RFID tracking, barcode scanning, IoT telemetry or a full enterprise resource planning platform. These may be considered later after the core inventory database has been implemented and tested.

7.0 Overview of the Proposed System

The proposed system will be a browser accessible application connected to a central relational database. Authorized users will use the system to enter item records, record stock movement, search for inventory information, check item status and generate reports. The application will operate through the local intranet during the initial deployment.

7.1 Main User Roles

The Administrator will manage users, roles, departments, categories and system settings. The Store Manager will manage day to day inventory entries, suppliers and stock transactions. Staff users will view authorized inventory information and make inventory related requests or confirmations based on the final workflow approved by the ICT team.

7.2 Main System Modules

- Authentication and role based authorization module
- Inventory catalogue and asset registration module
- Stock transaction and stock balance module
- Supplier management module
- Dashboard, search and report generation module
- Audit trail and security monitoring module

8.0 Proposed Methodology and Development Approach

8.1 Requirements Gathering

The project will begin with requirements gathering because the database must reflect the real inventory problem. Information will be collected through the supervisor brief, review of available documents, observation of inventory related activities and discussion with relevant users where permission is granted [1].

8.2 Database Design Approach

The database will be designed in three stages: conceptual design, logical design and physical design. The conceptual design will identify the main entities and relationships. The logical design will convert them into normalized relational tables with primary keys and foreign keys. The physical design will define MySQL tables, data types, constraints and indexes. This approach follows the standard database design methodology of moving from requirements analysis to conceptual, logical and physical design [5].

The design will be normalized to at least Third Normal Form. This will reduce unnecessary duplication, prevent update anomalies and support reliable reports. Where the real world requires it, the design may use associative tables to correctly represent many to many relationships such as item suppliers, stock transactions and asset assignments.

8.3 Development Approach

The system will be developed incrementally. The first increment will handle login, roles and master data. The second will handle item registration, suppliers and categories. The third will handle stock transactions and balances. The fourth will focus on dashboard reports, audit logs, testing and documentation.

8.4 Testing and Validation

Testing will cover database constraints, login and access rights, item registration, stock received, stock issued, stock balance accuracy, low stock alerts, report filtering, search functions and audit log recording. Security testing will include checking for invalid input, unauthorized access attempts and session related weaknesses.

9.0 Proposed Technology Stack

The proposed development environment will use open and locally available tools. The backend may be implemented using PHP with MySQL or MariaDB through XAMPP during development. The frontend will use HTML, CSS, Bootstrap and JavaScript. Chart.js may be used for simple dashboard charts. Database access will use parameterized queries to reduce SQL injection risks [1].

The project brief permits suitable frameworks such as Laravel, Django, ASP.NET and Node.js. For the internship prototype, PHP and MySQL remain practical because they are easy to deploy locally, easy to demonstrate and sufficient for a clear database focused implementation [1].

10.0 Significance of the Study

The proposed system will support better visibility of institutional resources. Management will be able to know available items, faulty assets, stock levels, departmental allocations and supplier related records. This can improve budgeting, reduce duplicate procurement and support faster decision making.

For the ICT department, the system will simplify the tracking of computers, network devices, projectors, printers and other technical assets. For stores and administration, it will improve stock records and issue history. For research and engineering units, it will help track tools, equipment and consumables needed for practical work.

11.0 Ethical Considerations and Institutional Permissions

The project will respect UIRI institutional procedures and confidentiality. Real inventory data, staff data and supplier information will only be used with permission from authorized personnel. Where real data is sensitive or unavailable, dummy records will be used for demonstration and testing.

User passwords will not be stored in plain text. User actions affecting important records will be logged. The system will be designed to avoid exposing internal records to unauthorized users or external systems.

12.0 Project Limitations

The project may be limited by the internship timeframe, availability of complete historical inventory records and access to users for validation. Some manual inventory records may also require cleaning before they can be entered into the system.

The initial version will focus on the core database and web application functions. Advanced automation such as barcode scanning, RFID and IoT based equipment monitoring will be left as future enhancements.

14.0 Proposed Work Plan

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AV	AW	AX		
1	URI INVENTORY MANAGEMENT SYSTEM DATABASE PROJECT																																																	
2	Makerere University		Project Start:		Jun 15, 2026																																													
3	Database Project Team		Today:		Jun 28, 2026																																													
4	By Keith, Tracy & Deo		Display Week:		1																																													
5																																																		
6	TASK	ASSIGNED TO	PROGRESS	START	END	DAYS	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S		
7																																																		
8	Phase 1: Project Initiation and Requirements																																																	
9	Supervisor briefing and scope confirmation	Keith, Tracy & Deo	100%	6/15/26	6/15/26	1																																												
10	URI inventory process study	Keith & Tracy	100%	6/15/26	6/17/26	3																																												
11	Stakeholder and user requirements gathering	Tracy & Deo	100%	6/16/26	6/19/26	4																																												
12	Review of project brief and research files	Keith	100%	6/17/26	6/19/26	3																																												
13	Proposal restructuring and approval draft	Keith & Tracy	100%	6/18/26	6/21/26	4																																												
14	Phase 2: Requirements Analysis and SRS																																																	
15	Problem analysis and business rules	Keith, Tracy & Deo	100%	6/22/26	6/23/26	2																																												
16	SRS preparation	Tracy	100%	6/22/26	6/26/26	5																																												
17	Use case identification	Deo & Tracy	100%	6/24/26	6/27/26	4																																												
18	Inventory categories and department mapping	Keith & Deo	100%	6/23/26	6/27/26	5																																												
19	Security requirements and audit review	Keith	60%	6/27/26	6/28/26	2																																												
20	Phase 3: Database Design and Documentation																																																	
21	Entity identification and relationship mapping	Keith	0%	6/29/26	6/30/26	2																																												
22	Data dictionary development	Tracy & Deo	0%	6/30/26	7/2/26	3																																												
23	Conceptual ERD preparation	Keith & Deo	0%	6/30/26	7/3/26	4																																												
24	Logical ERD and 3NF normalization	Keith	0%	7/2/26	7/4/26	3																																												

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15.0 Estimated Budget

Resource Component	Specification	Estimated Cost (UGX)
Development Workstation	Laptop or desktop computer for coding, documentation and database testing	2,200,000
XAMPP Server Suite	Open-source Apache, PHP and MySQL development bundle	0
Internet and Research Data	Two-month development and documentation support	300,000
Report Production	Printing and binding of project documentation	120,000
Backup Storage and Contingency	External backup, stationery and minor deployment support	200,000
Total Estimate	Estimated prototype project cost	2,820,000

16.0 Conclusion

This proposal has presented the need, scope, objectives, methodology and expected deliverables for a web based Inventory Management System for Uganda Industrial Research Institute. The proposed system responds to the practical need for better inventory visibility, stock movement tracking, asset accountability and management reporting across UIRI operations.

The next stage will convert this proposal into a detailed Software Requirements Specification, followed by data dictionary preparation, conceptual and logical ERD design, normalization, UML diagrams, implementation, testing, user documentation and final demonstration. The SRS will follow a structured requirements format consistent with IEEE software requirements documentation guidance [6].

17.0 References

- [1] ICT Team, UIRI Namanve, "Internship Project Question: Design and Development of an Inventory Management System for Uganda Industrial Research Institute (UIRI)," Uganda Industrial Research Institute, Namanve, Uganda, Project Brief, 2026.
- [2] Uganda Industrial Research Institute, "Uganda Industrial Research Institute: Home," UIRI Official Website. Accessed: Jun. 28, 2026. [Online]. Available: <https://uiri.go.ug/>
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